Rafael Velecela

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Education

University of South Florida, Tampa, FL
Bachelor of Science in Computer Science, Expected May 2027
Cumulative GPA: 3.75/4.00
Relevant Coursework: Trustworthy Infrastructure, Software Engineering, Algorithms

Experience

RAG Documentation Retrieval
Developer, Personal Project; October 2026–Present

- Build a FastAPI semantic-search endpoint using Qwen embeddings and Qdrant to retrieve relevant documentation by question.
- Configure two Docker Compose services with local ports, persistent storage, and read-only mounts to limit access.

InCollege
Developer, Tester, and Scrum Master, Team Kansas; September 2026–Present

- Implement COBOL file I/O and duplicate checks to retain up to 50 job applications per user across restarts.
- Document seven sprints of COBOL features, file organization, and test procedures to help teammates run and verify the application.
- Coordinate development and regression testing in a five-person Scrum team to verify data persistence and input validation.

Artemis: Offline AI Voice Assistant
Developer, Personal Project; April 2026–Present

- Set up and configure Raspberry Pi and ESP32 hardware with least-privilege MQTT access to support local sensor monitoring.
- Automate Mosquitto provisioning and systemd service startup to make configuration and boot processes repeatable.
- Organize technical documentation for account management, device setup, and troubleshooting to make procedures easier to follow.
- Build an 18-prompt benchmark to evaluate local models for response accuracy, latency, and token throughput.
- Run detail-oriented regression checks after updates to catch broken functionality; develop unit and integration tests to automate checks.

Skills

Systems and Tools: Ubuntu/WSL2, Raspberry Pi, ESP32, Docker, Docker Compose, Git
Automation: Python, Bash, systemd services, MQTT account provisioning and access tests
Applications and Protocols: MQTT, Mosquitto, FastAPI, Qdrant
Programming: Python, C, COBOL